Uber Rides Analysis Dashboard

Case Study | Power BI, DAX, Data Modeling

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An interactive Power BI dashboard that analyzes ~150,000 Uber ride bookings to understand rider behavior, revenue, vehicle performance, demand patterns, and customer ratings.

1. Project Summary

This project turns a raw ride-bookings dataset into a six-page interactive Power BI report. It helps a business team answer, at a glance, how many rides are completed or lost, which vehicles and payment methods bring in the most revenue, who the riders are, why rides get canceled, and when and where demand peaks.

Item	Details
Goal	Monitor bookings, revenue, riders, vehicles, and locations in one place
Dataset	Uber ride bookings, about 150,000 rows
Tools	Power BI Desktop, DAX, data modeling (date table, relationships)
Deliverable	6-page interactive report with vehicle slicer, month/quarter toggle, and page navigation

2. Business Questions

- How many bookings are completed vs. lost, and how much revenue do they generate?
- Which vehicle types and payment methods contribute the most revenue?
- Who are the riders: first-time, returning, or regular?
- Why do rides get canceled, and who cancels (customer or driver)?
- Which time slots, days, and areas are the busiest?
- How do riders and drivers rate the service?

3. Approach

Data preparation and modeling

- Loaded the bookings table and a separate image table (vehicle icons) and related them to the main table by vehicle.
- Built a calendar table to support monthly and quarterly analysis, with index columns so months, quarters, and weekdays sort correctly.
- Created a measures table for KPIs: completed bookings, booking count, revenue, total distance, average distance, and ratings.
- Added calculated columns such as **Time Slot** (3-hour windows derived from the booking time) and **Time Slot Sort**.

Report design

- A consistent layout on every page: left navigation menu, vehicle icon slicer, KPI cards, and a vehicle highlight panel.
- A Month / Quarter toggle built with bookmarks, and a Complete / Cancelled / Incomplete explanation panel.
- Black, white, and grey styling so the data stands out. Custom icons and images for navigation and filtering.

4. Headline KPIs

KPI	Value	Comment
Completed bookings	93K (92,551)	About 62% of all bookings
Lost bookings	About 57K (56,817)	About 38% of all bookings
Total bookings	149K	
Revenue	52M (51.8M)	About 560 per completed booking
Total distance	3M km	Average of about 24.6 km per ride
Unique customers	104K	
Average customer rating	4.40	
Average driver rating	4.23	

5. Dashboard Pages

Page	Purpose	Main visuals
Home	Landing page and navigation	Title, description, page buttons
Overview	Business summary	KPIs, monthly and quarterly completed bookings and revenue, revenue by vehicle, top pickup and drop, ratings
Vehicle	Vehicle performance	Table with customers, revenue, completed bookings, contribution %, and monthly sparkline
Revenue	Where money comes from	Revenue by month/quarter, vehicle, payment method, and customer
Rider	Rider behavior	First-time / return / regular riders, customers by payment method, cancellations by reason, customer detail table
Location	Demand and distance	Monthly distance, distance by vehicle, time slot by weekday heatmap, bookings by location

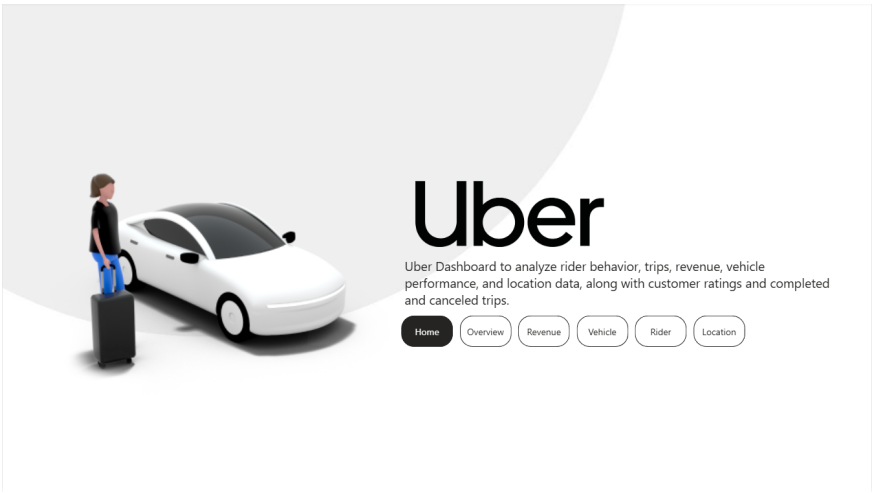


Figure 1. Home page with navigation.

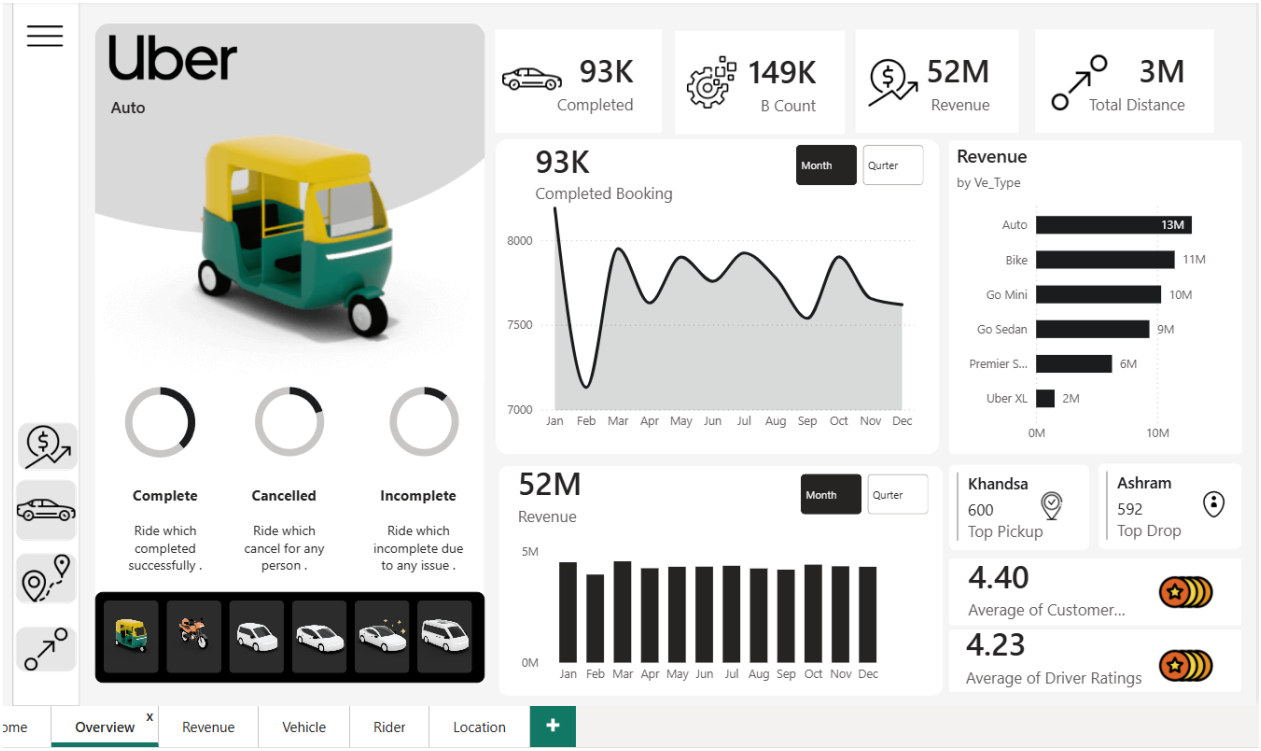


Figure 2. Overview page: KPIs, monthly trends, revenue by vehicle, top locations, and ratings.

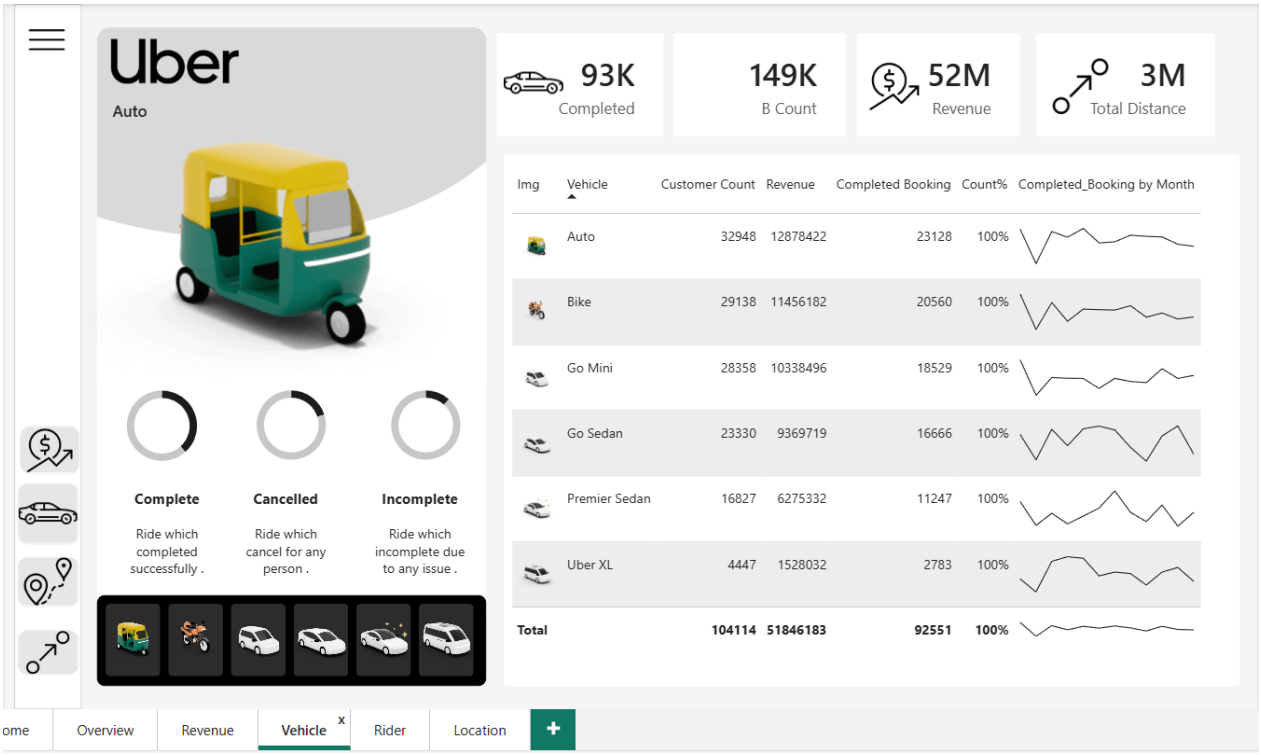


Figure 3. Vehicle page: customers, revenue, completed bookings, contribution, and monthly trend per vehicle.

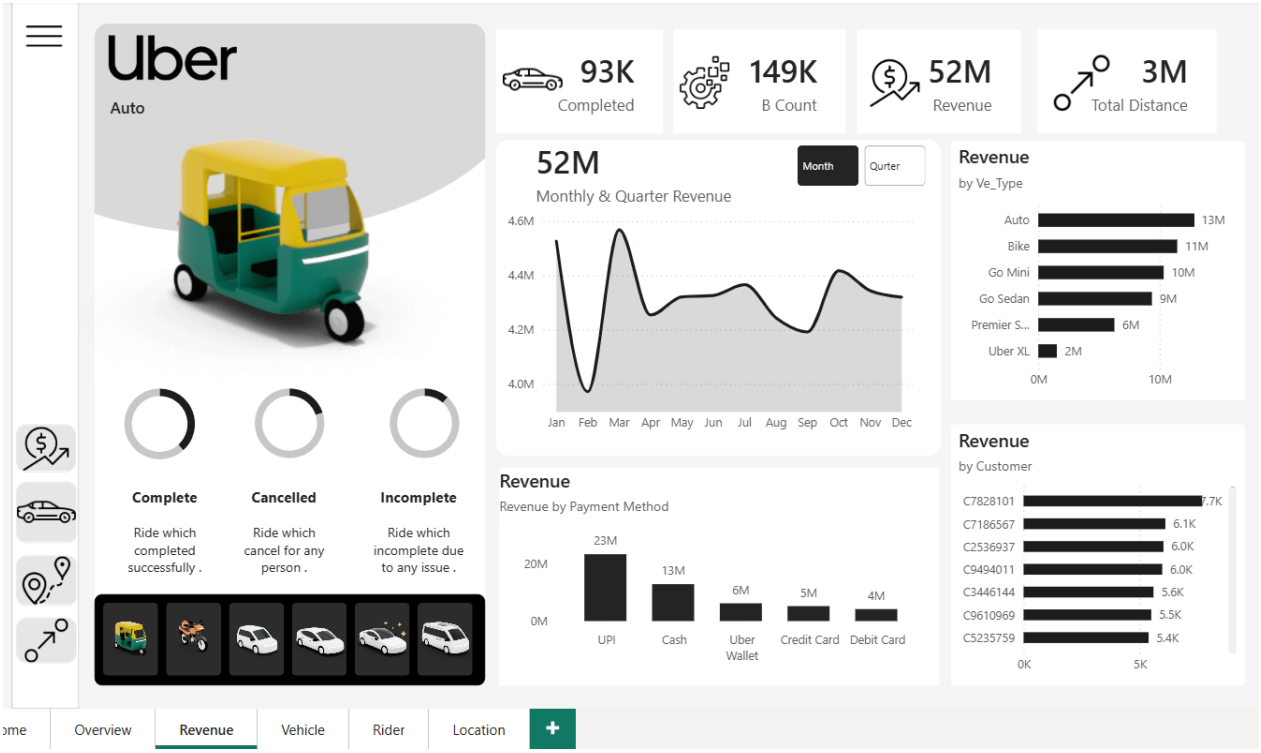


Figure 4. Revenue page: revenue by month/quarter, vehicle type, payment method, and customer.

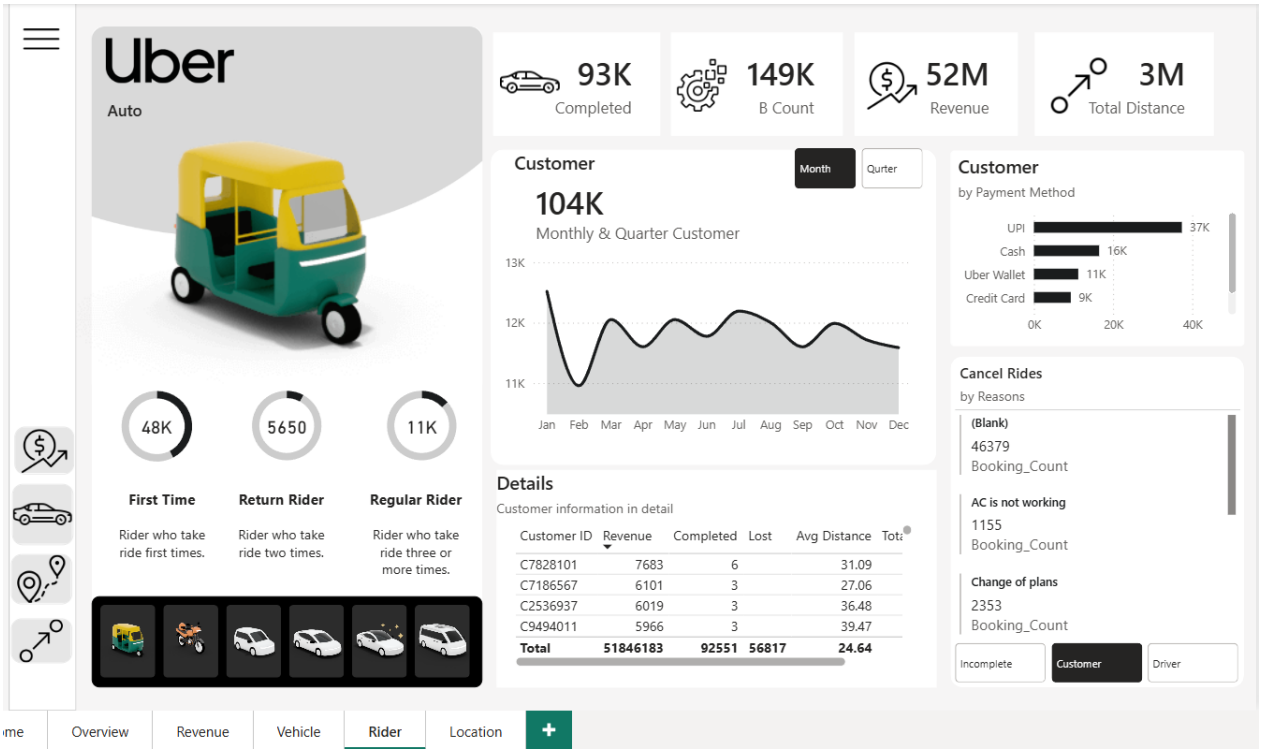


Figure 5. Rider page: rider types, payment methods, cancellation reasons, and customer details.

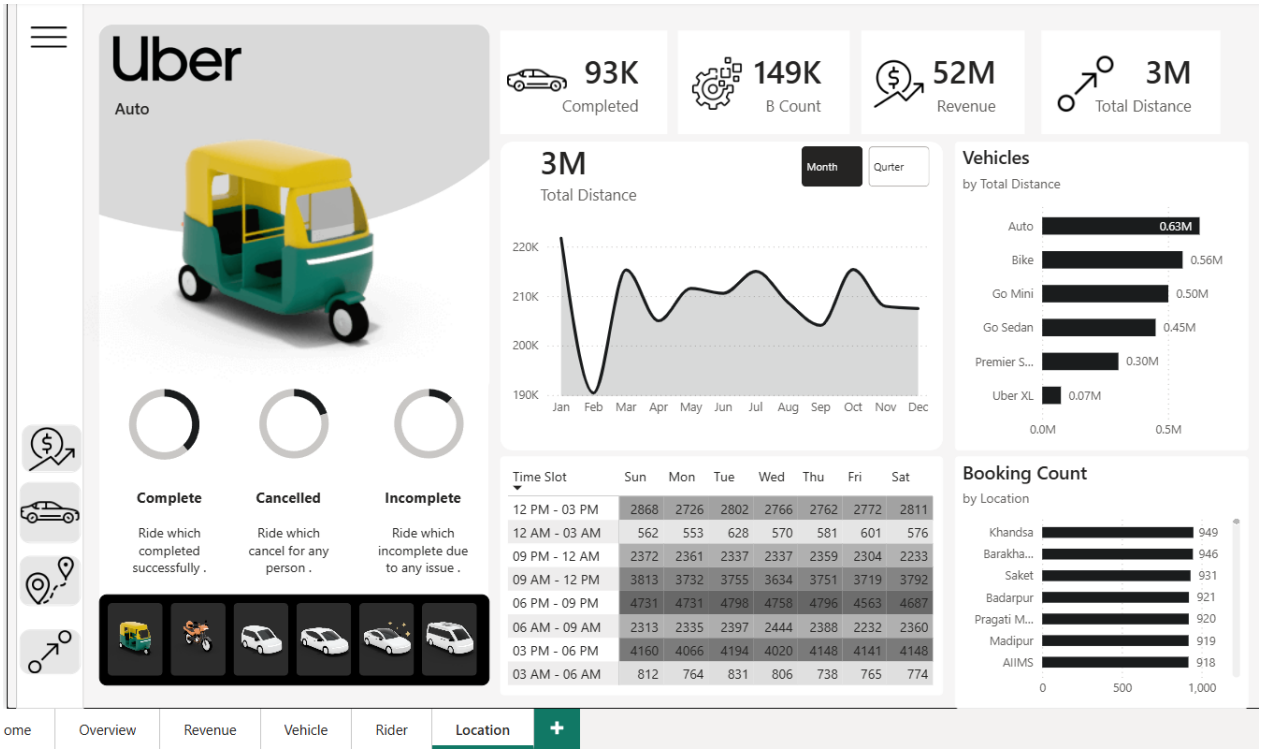


Figure 6. Location page: distance trends, time slot heatmap, and bookings by location.

6. Key Insights

Revenue and vehicles

- **Auto, Bike, and Go Mini bring in about two thirds of revenue** (13M, 11M, and 10M out of 52M). Go Sedan adds 9M, Premier Sedan 6M, and Uber XL only 2M.
- **Uber XL is the smallest segment**: about 4.4K customers and 2.8K completed bookings, far below the other vehicles.
- Auto also covers the most distance (0.63M km), followed by Bike (0.56M) and Go Mini (0.50M).

Payments

- **UPI is the leading payment method** with 23M in revenue (about 44% of the total), followed by Cash (13M) and Uber Wallet (6M). Cards together account for about 9M.

Demand patterns

- **The evening is the peak**: the 06 PM to 09 PM slot has the highest bookings on every weekday (about 4,600 to 4,800 per weekday), followed by 03 PM to 06 PM and 09 AM to 12 PM.
- Late night is the quietest period: 12 AM to 03 AM has only about 550 to 630 bookings per weekday.
- Demand is **steady across weekdays**; the time of day matters far more than the day of the week.
- **February is the lowest month** in completed bookings, revenue, distance, and customers, then recovers in March.
- Bookings are spread evenly across areas: the top locations (Khandsa, Barakhamba, Saket) are within a narrow range of about 920 to 950 bookings.

Cancellations and ratings

- **38% of bookings are lost**, which is the biggest improvement opportunity in the data.
- A large number of canceled rides have **no recorded reason (Blank)**, which limits root-cause analysis. Among recorded reasons, examples include "Change of plans" and "AC is not working".
- Ratings are healthy: 4.40 for customers and 4.23 for drivers.

7. Recommendations

These are suggestions based on the patterns above, and should be validated with additional data.

- **Fix the cancellation reason capture.** Make the reason mandatory so the business can find out why rides are lost.
- **Plan driver supply for 06 PM to 09 PM**, with incentives in peak slots to reduce lost bookings.
- **Prioritize Auto, Bike, and Go Mini** for promotions and availability, as they drive most of the revenue.
- **Review the Uber XL offering**: increase awareness or reconsider coverage given its low volume.
- **Investigate the February dip**: check seasonality, pricing, or supply issues.
- **Encourage digital payments**: UPI is already dominant, so targeted offers could move cash users over.

8. Challenges and Solutions

Challenge	Solution
Time slots were sorting alphabetically, giving an illogical order (for example 03 PM before 06 AM)	Created a **Time Slot Sort** column and used Sort by column
Circular dependency error between Time Slot and Time Slot Sort	Calculated both columns directly from the booking **hour** instead of one depending on the other
Contribution % showed 100% for every vehicle	Used **DIVIDE** with **CALCULATE + ALLSELECTED** on the vehicle column used in the table
Showing vehicle images inside visuals	Stored image links in a separate table and set the Data category to Image URL

9. Limitations and Next Steps

- The analysis is descriptive. Forecasting demand and revenue would be a useful next step.
- Currency is not specified in the data, so revenue is shown as raw values.
- Add average wait time, surge pricing, and driver-level metrics if the data becomes available.
- Publish the report online and add row-level filters for different user roles.

10. Skills Demonstrated

- Dashboard design and storytelling in Power BI
- DAX: calculated columns, measures, ALLSELECTED, SWITCH, time intelligence basics
- Data modeling: date table, relationships, image tables
- Business analysis: turning data into insights and recommendations